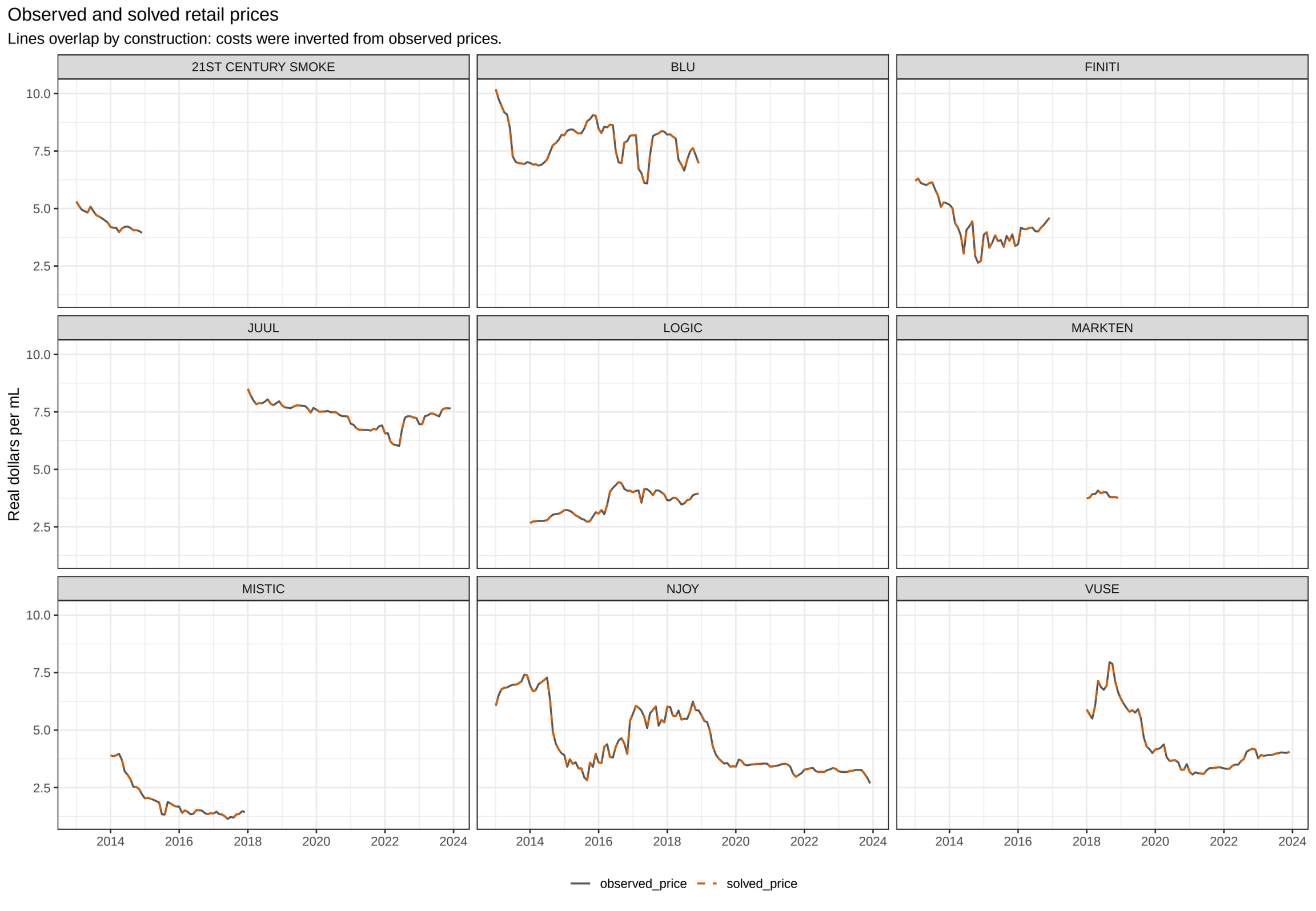

Observed and solved retail prices
Lines overlap by construction: costs were inverted from observed prices.
21ST CENTURY SMOKE
BLU
FINITI
JUUL
LOGIC
MARKTEN
MISTIC
NJOY
VUSE
Real dollars per mL
10.0
7.5
5.0
2.5
2014
2016
2018
2020
2022
2024
observed_price
solved_price

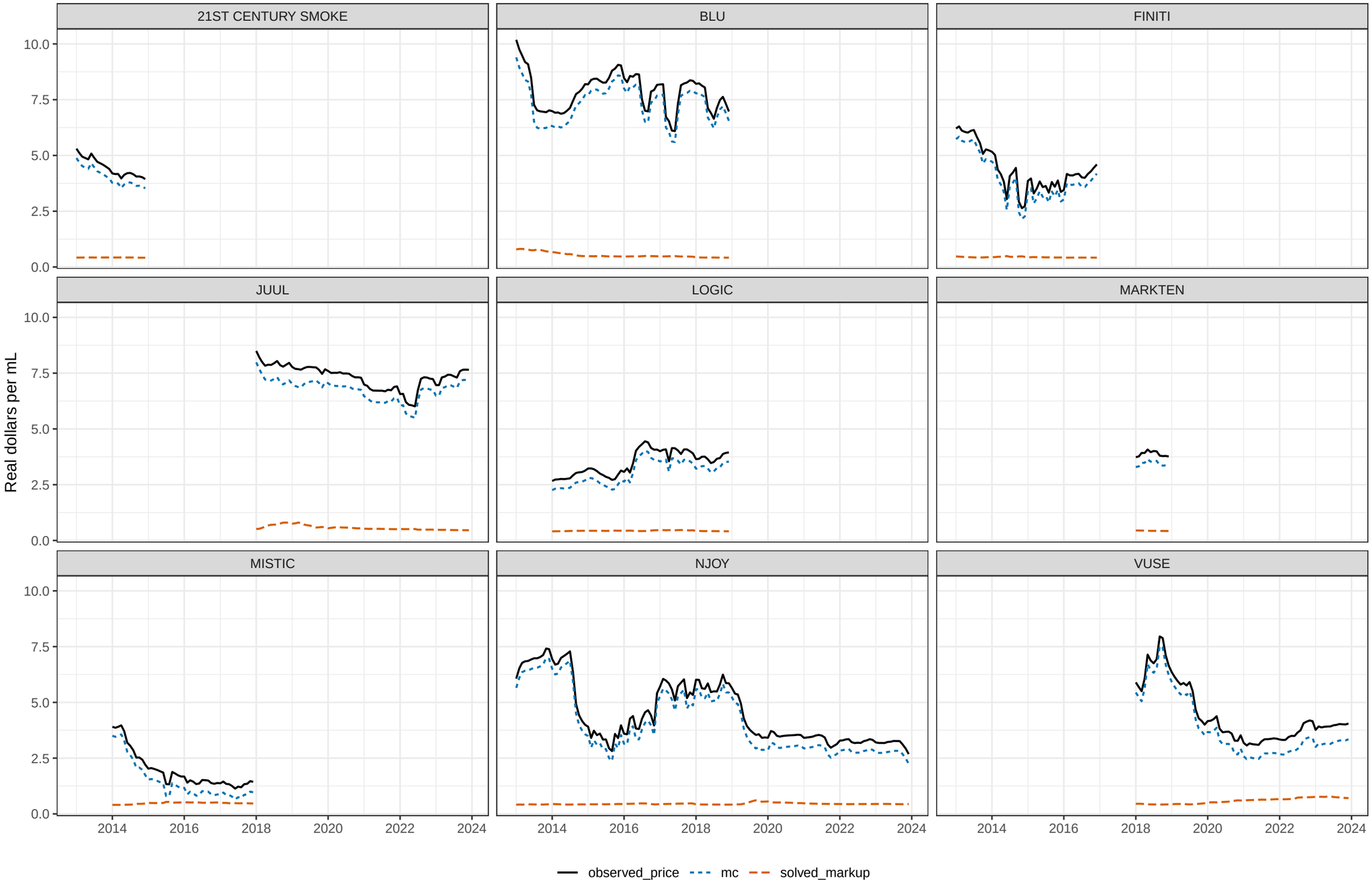
Solved minus observed price

Separate error axis exposes differences hidden by overlapping price lines.



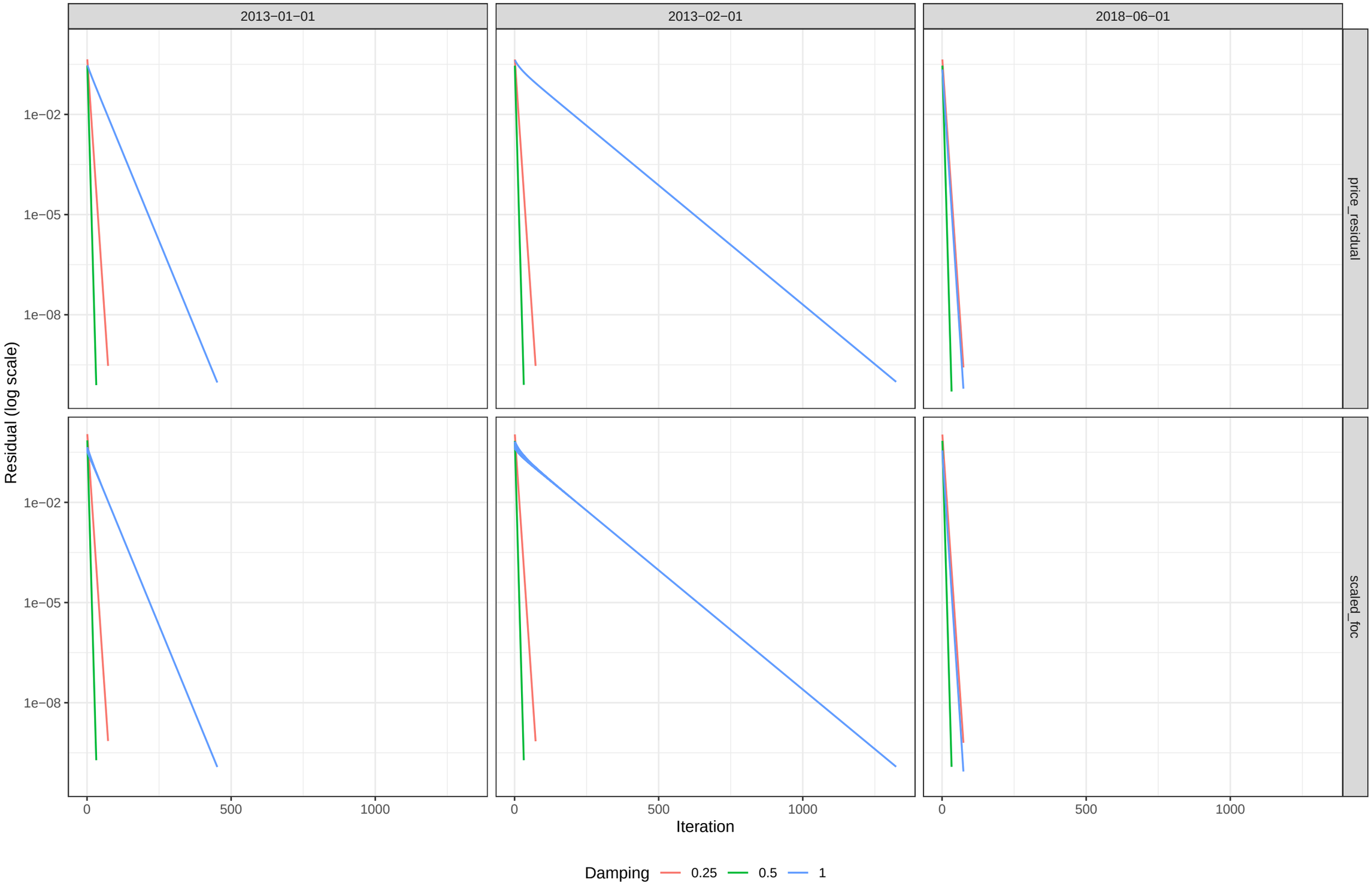
Price, recovered marginal cost, and markup

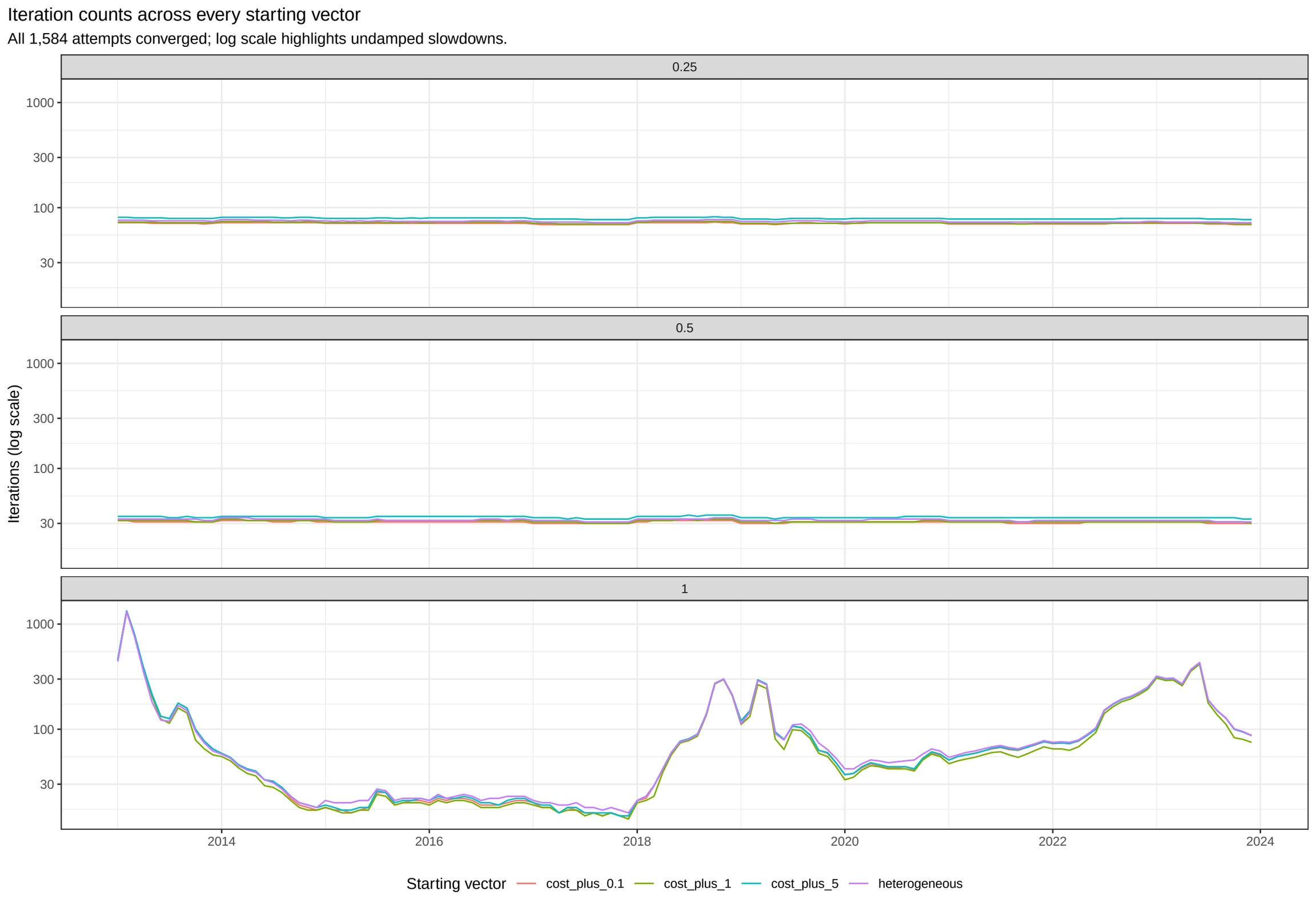
Integrated−channel costs; profits exclude unspecified fixed operating costs.



Convergence from marginal cost + $1/mL

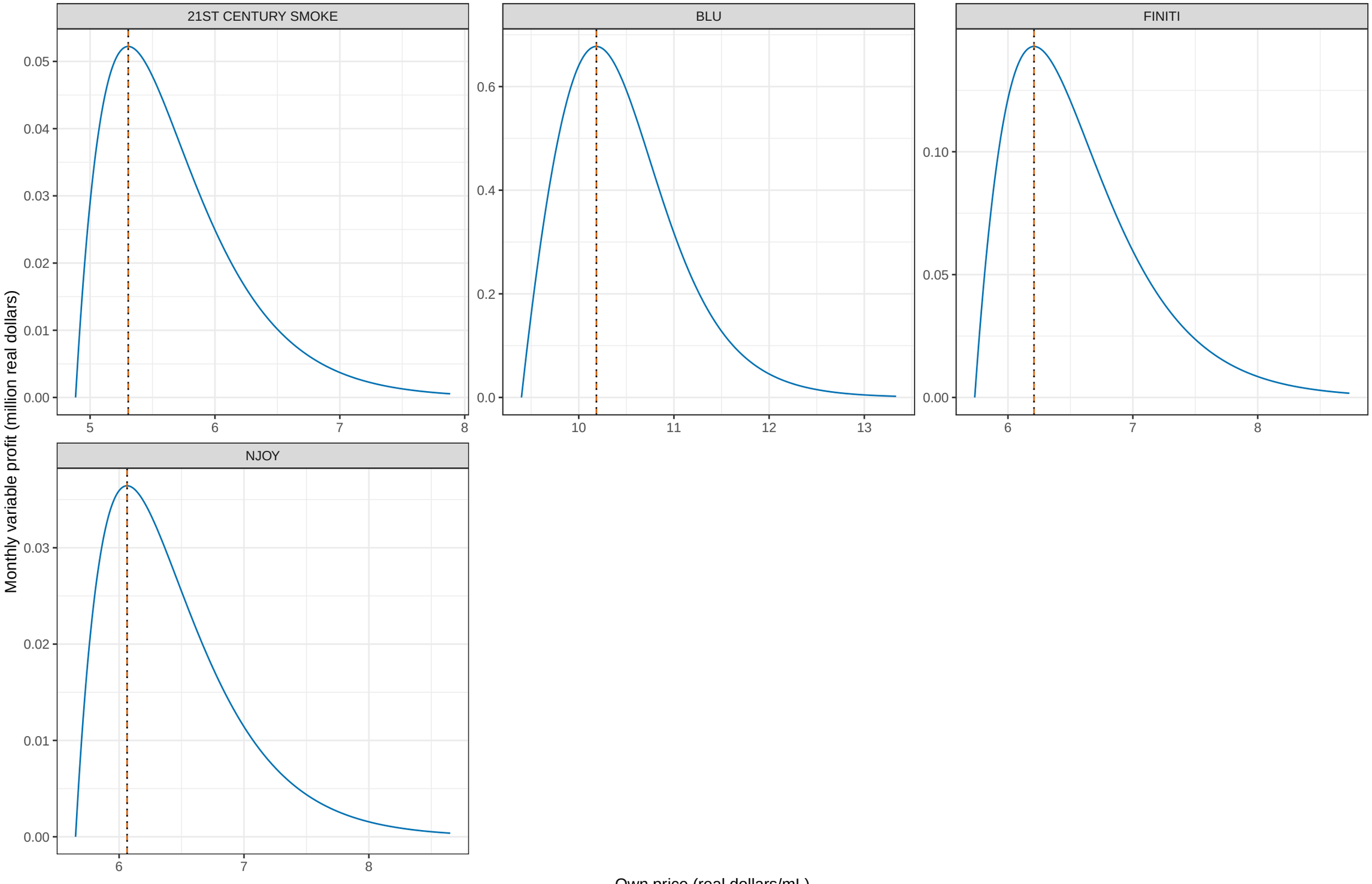
First, middle, and slowest month. Full undamped iteration path retained.




Iteration counts across every starting vector
All 1,584 attempts converged; log scale highlights undamped slowdowns.
0.25
0.5
1
Iterations (log scale)
1000
300
100
30
2014
2016
2018
2020
2022
2024
Starting vector
cost_plus_0.1
cost_plus_1
cost_plus_5
heterogeneous

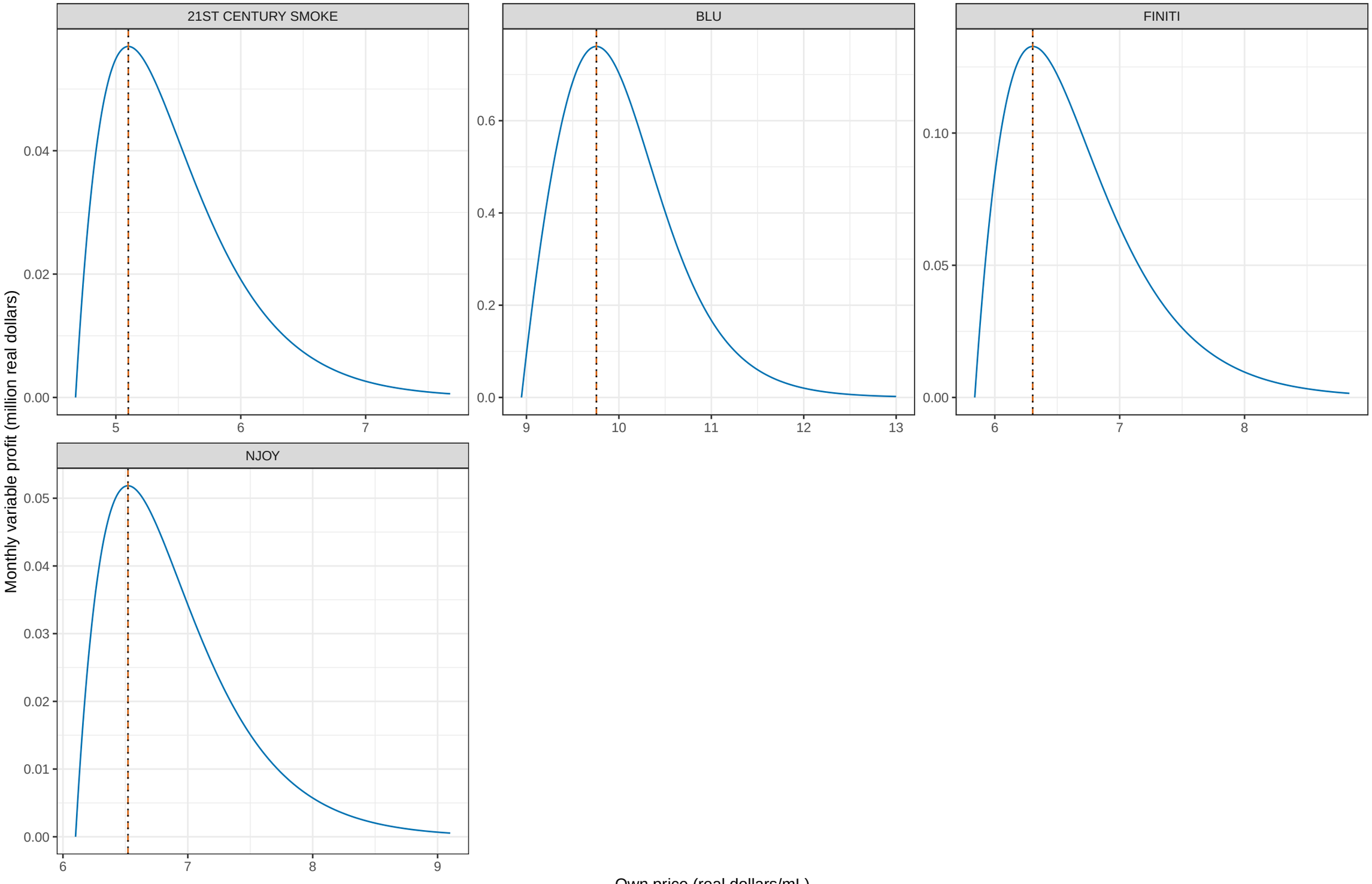
Unilateral variable−profit curves: 2013−01−01

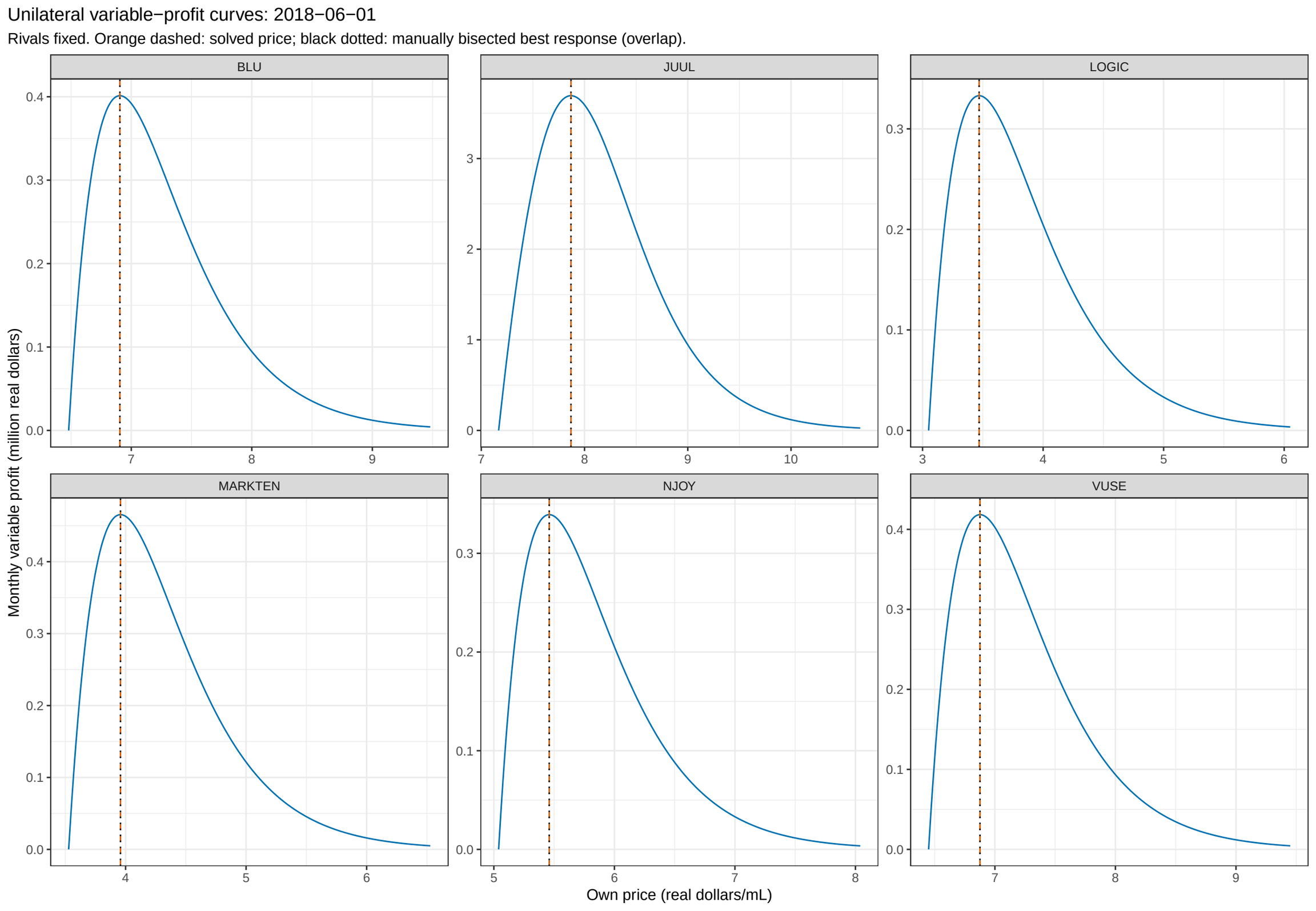
Rivals fixed. Orange dashed: solved price; black dotted: manually bisected best response (overlap).



Unilateral variable−profit curves: 2013−02−01

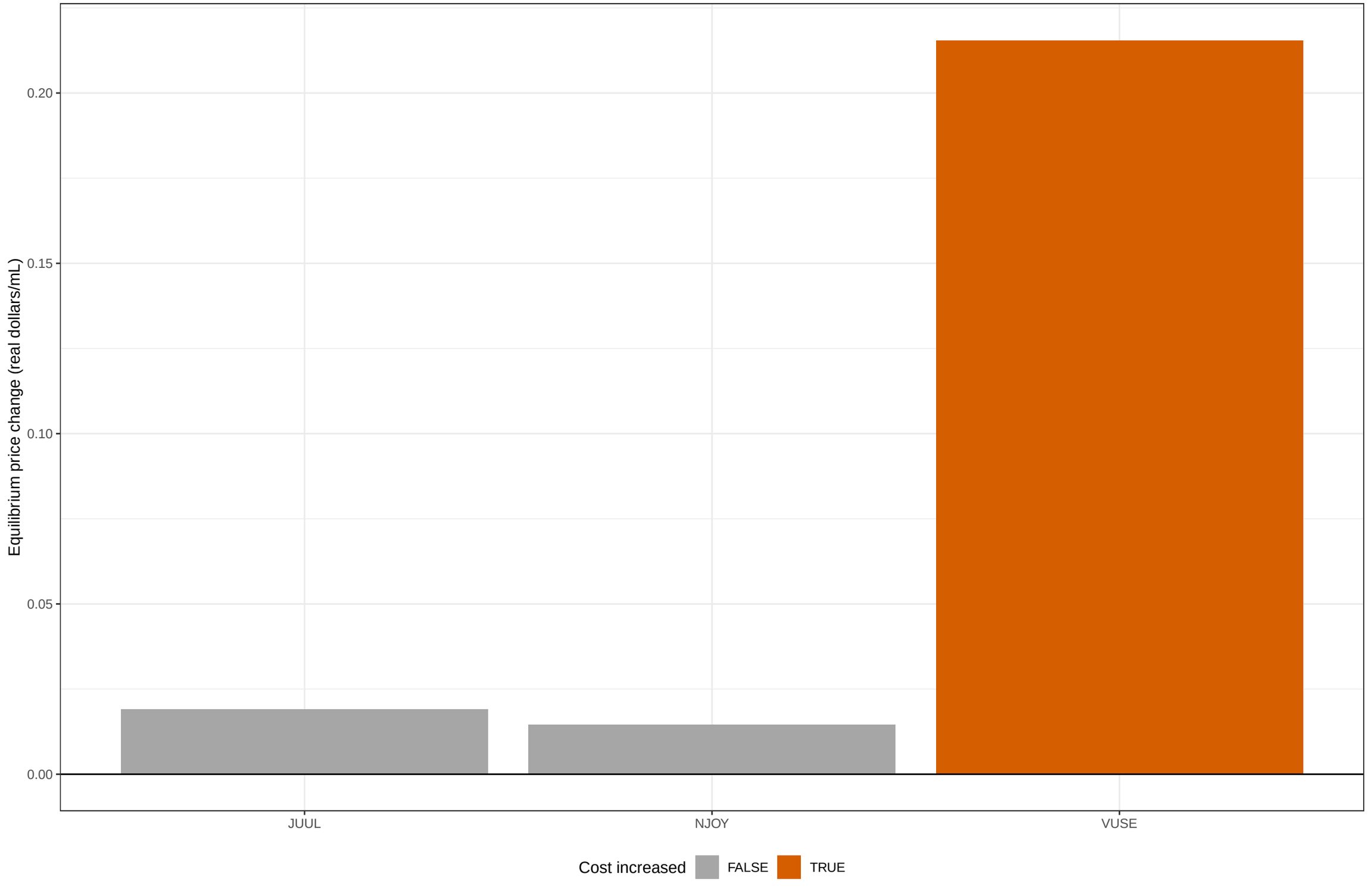
Rivals fixed. Orange dashed: solved price; black dotted: manually bisected best response (overlap).

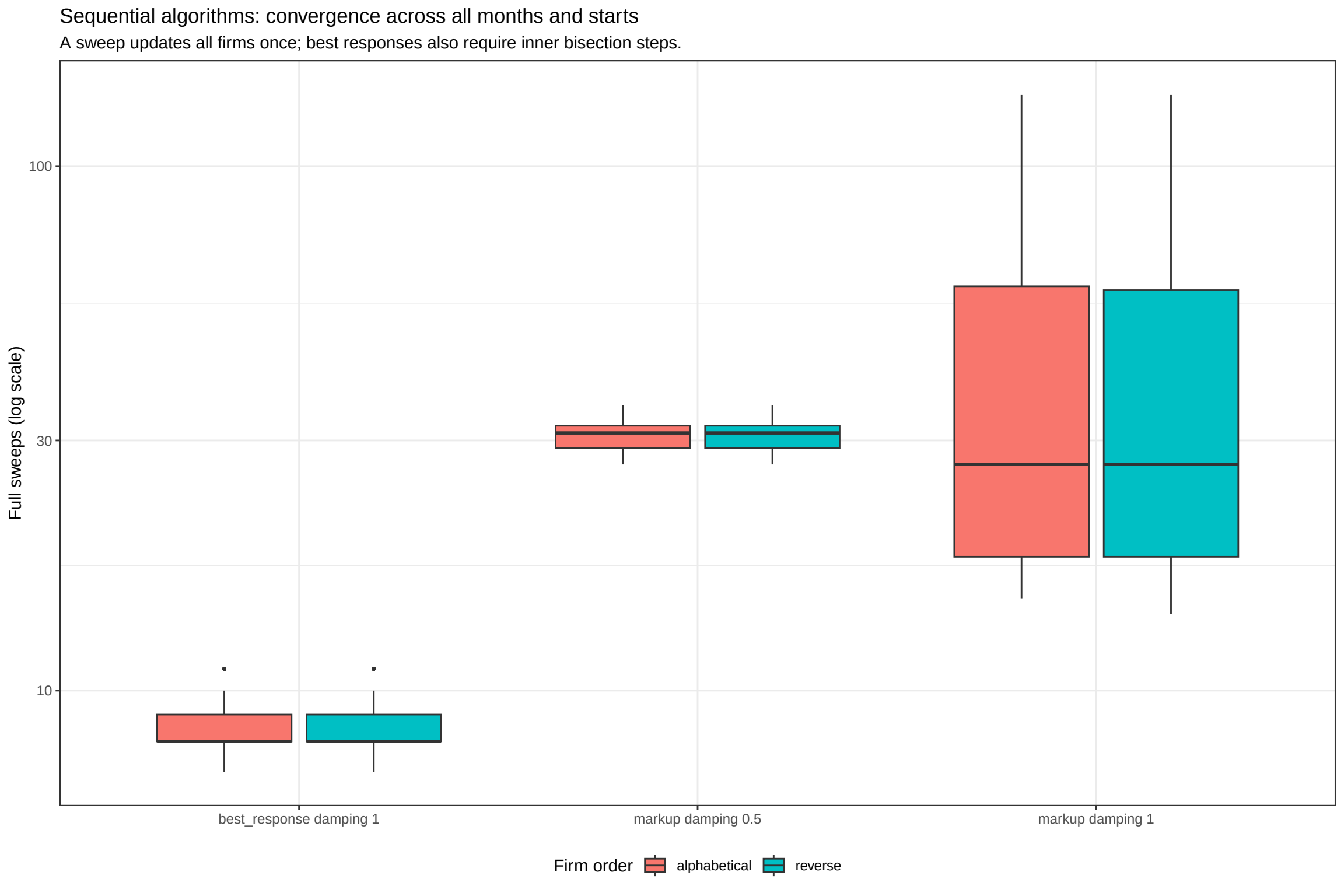


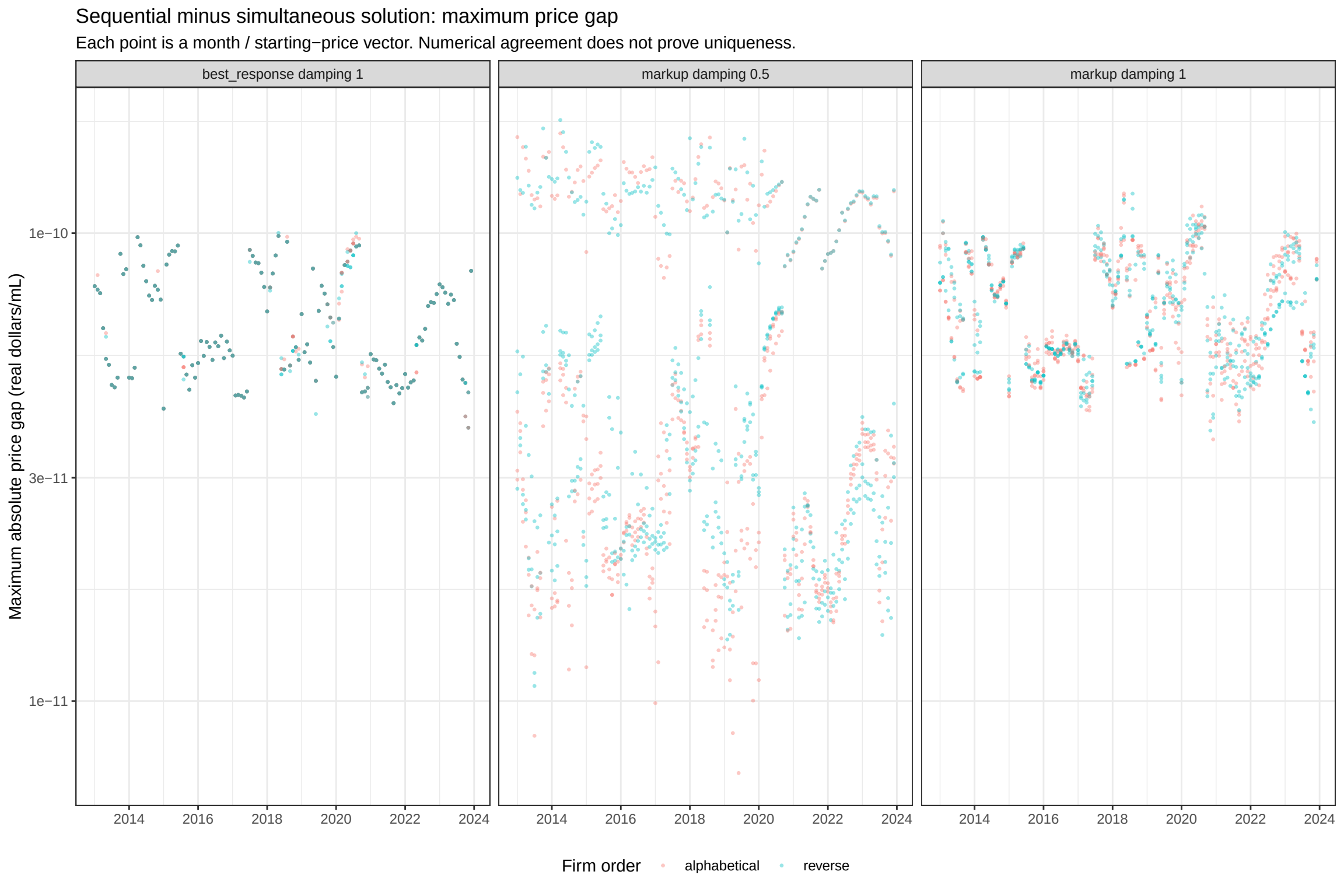

Unilateral variable−profit curves: 2018−06−01
Rivals fixed. Orange dashed: solved price; black dotted: manually bisected best response (overlap).
BLU
JUUL
LOGIC
MARKTEN
NJOY
VUSE
Monthly variable profit (million real dollars)
Own price (real dollars/mL)

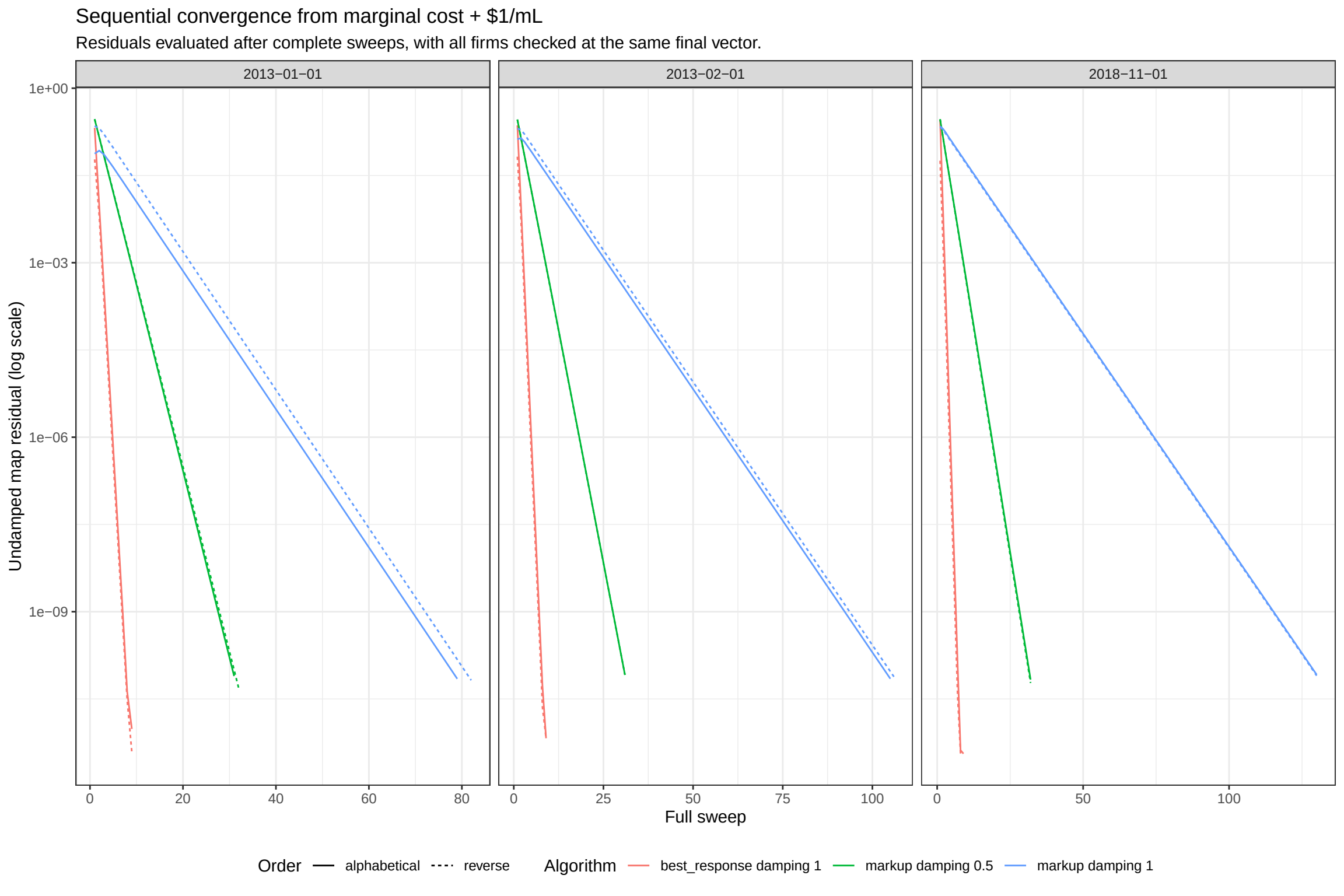
Numerical cost perturbation: VUSE 2023−12

Largest−volume firm in latest month: effective marginal cost +10%; demand states and fringe utility fixed.




Sequential algorithms: convergence across all months and starts
A sweep updates all firms once; best responses also require inner bisection steps.
Full sweeps (log scale)
100
30
10
best_response damping 1
markup damping 0.5
markup damping 1
Firm order
alphabetical
reverse


Sequential minus simultaneous solution: maximum price gap
Each point is a month / starting−price vector. Numerical agreement does not prove uniqueness.
best_response damping 1
markup damping 0.5
markup damping 1
Maximum absolute price gap (real dollars/mL)
1e−10
3e−11
1e−11
2014
2016
2018
2020
2022
2024
Firm order
alphabetical
reverse


Sequential convergence from marginal cost + $1/mL
Residuals evaluated after complete sweeps, with all firms checked at the same final vector.
2013−01−01
2013−02−01
2018−11−01
Undamped map residual (log scale)
1e+00
1e−03
1e−06
1e−09
0
20
40
60
80
0
25
50
75
100
0
50
100
Full sweep
Order
alphabetical
reverse
Algorithm
best_response damping 1
markup damping 0.5
markup damping 1